

DOUGLAS RIVER, Australia

WMO: 941280

Lat: 13.8344S

Lon: 131.1872E

Elev: 44

StdP: 100.80

Time Zone: 9.50 (AUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	10.5	12.1	-4.5	2.6	26.8	-1.7	3.3	25.0	9.1	27.3	8.5	26.8	1.2	140	0.351

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	13.8	38.0	22.2	37.1	22.6	36.3	23.0	29.4	32.3	27.6	32.2	27.0	31.8	3.3	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.8	25.4	31.6	26.5	22.1	29.6	25.8	21.2	28.7	97.4	32.3	88.1	32.3	85.4	31.9	36.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.8	6.8	5.9	DB	7.4	40.0	1.6	0.9	6.2	40.7	5.3	41.3	4.5	41.8	3.3	42.4	(4)
(5)			WB	5.6	32.0	1.1	3.4	4.8	34.5	4.2	36.5	3.5	38.4	2.8	40.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.0	28.4	28.3	28.4	27.6	25.3	22.8	22.7	24.0	27.8	29.9	29.8	29.2	(6)	
	DBStd	3.13	1.45	1.36	1.13	1.52	2.37	2.70	2.34	2.22	1.95	1.53	1.41	1.48	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	3	0	0	0	0	0	2	1	0	0	0	0	0	(9)	
	CDD10.0	6208	571	511	571	528	474	383	395	433	534	617	594	595	(10)	
	CDD18.3	3169	312	278	313	278	216	135	138	175	284	359	344	337	(11)	
	CDH23.3	36067	3221	2782	3254	3019	2269	1438	1612	2206	3647	4686	4151	3783	(12)	
	CDH26.7	17817	1329	1117	1459	1522	1092	599	706	1123	2120	2729	2229	1792	(13)	
(14) Wind	WSAvg	2.1	1.8	1.7	1.6	2.1	2.6	3.0	2.8	2.4	2.1	2.1	1.8	1.7	(14)	
(15) Precipitation	PrecAvg	1300	316	289	192	46	13	0	1	2	5	39	128	269	(15)	
	PrecMax	1814	703	581	575	380	111	2	12	23	59	150	228	598	(16)	
	PrecMin	925	68	121	3	0	0	0	0	0	0	3	36	95	(17)	
	PrecStd	243	153	125	143	78	25	0	2	5	11	32	49	123	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.1	35.7	35.9	36.2	35.3	34.1	33.7	35.7	38.0	39.6	38.9	37.9	(19)	
		MCWB	26.0	26.1	24.9	23.2	21.8	21.2	19.9	20.7	20.9	21.0	22.9	24.9	(20)	
	2%	DB	34.7	34.6	34.9	35.2	34.2	32.7	32.6	34.3	37.0	38.3	37.6	36.2	(21)	
		MCWB	26.4	26.0	25.0	23.4	21.6	20.6	19.5	20.5	21.0	21.6	23.5	25.6	(22)	
	5%	DB	33.8	33.7	34.1	34.4	33.3	31.6	31.7	33.2	36.1	37.4	36.5	35.1	(23)	
		MCWB	26.4	26.1	25.1	23.3	21.3	20.1	19.0	19.4	20.9	22.0	24.2	25.7	(24)	
	10%	DB	32.6	32.6	33.2	33.5	32.2	30.3	30.7	32.2	35.3	36.4	35.3	34.0	(25)	
		MCWB	26.5	26.0	25.3	23.2	21.1	19.3	18.4	18.7	20.9	22.3	24.7	26.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.2	27.9	29.3	29.7	26.2	27.1	24.9	29.8	31.9	31.3	30.2	27.7	(27)	
		MCDB	32.1	32.6	32.0	32.9	30.0	28.3	29.2	31.2	32.8	35.0	33.8	33.4	(28)	
	2%	WB	28.0	27.2	27.3	26.6	25.1	23.0	22.3	23.1	24.7	26.1	27.6	27.2	(29)	
		MCDB	32.1	31.9	31.8	31.0	29.3	28.8	28.5	29.1	30.5	31.3	33.1	32.8	(30)	
	5%	WB	27.1	26.8	26.7	25.7	24.2	21.8	21.0	22.1	24.1	25.2	26.4	26.7	(31)	
		MCDB	31.5	31.3	31.2	30.0	28.6	28.6	28.1	28.7	29.5	30.4	31.6	32.1	(32)	
	10%	WB	26.6	26.4	26.3	25.2	23.3	20.7	19.9	21.1	23.5	24.7	25.8	26.4	(33)	
		MCDB	30.8	30.6	30.5	29.4	28.3	27.8	27.6	27.9	29.1	30.0	31.0	31.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	8.2	9.7	12.4	14.4	15.6	16.5	17.2	16.0	13.8	11.7	9.6	(35)	
		MCDBR	10.2	10.3	11.7	13.8	15.1	15.9	16.9	17.7	16.3	15.3	13.4	11.5	(36)	
	5% WB	MCWBR	3.4	3.3	3.5	3.9	4.2	5.3	5.7	5.9	4.6	3.9	3.3	3.2	(37)	
		MCWBR	8.9	8.8	9.4	10.6	11.3	14.4	15.1	15.6	14.0	12.8	11.2	9.9	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.417	0.409	0.395	0.374	0.369	0.343	0.339	0.357	0.424	0.452	0.462	0.427	(40)	
		taud	2.469	2.486	2.523	2.551	2.503	2.524	2.525	2.455	2.304	2.257	2.273	2.425	(41)	
	Ebn at Noon	Ebn at Noon	928	927	922	909	880	888	901	911	881	878	880	917	(42)	
		Edh at Noon	119	116	109	100	99	94	96	109	134	144	144	124	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	5.63	5.72	5.86	5.91	5.42	5.20	5.47	6.12	6.58	6.78	6.51	6.07	(44)	
	RadStd	RadStd	0.55	0.85	0.68	0.43	0.30	0.23	0.16	0.21	0.34	0.34	0.29	0.49	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	+0.33	N/A	N/A	+0.34	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page